

zkEVM Internship wrap-up

Scope

*take a look at combination of
guest program zkVM:*

ethrex $\leftrightarrow$ ZisK

Method

*measure AIR-costs of mainnet
block execution*

ZisK profiling is super helpful

Result

Combined speedup:

up to 42%

+ some side-quests

What works

01 apply hash precompiles correctly

- [Ethereum #6549](#) : use hash precompiles in eip-8025 hash tree root
- [Ethereum #6774](#) : route remaining hash calls through precompile

02 conditional compile things not applicable to zkVMs

- [Ethereum #6560](#) : remove rayon from eip-8025 execution path
- [Ethereum #6588](#) : remove sync from eip-8025 execution path

03 be more cautious about memory

- [Ethereum #6664](#) : resolve memory limit crash on zisk (*remove redundant allocations*)
- [Ethereum #6708](#) : replace index based jump targets with bit vector

04 apply zkVM specific instructions

- [Ethereum #6968](#) : utilize native u256 ZisK arithmetic

Side-quests

[Ethrex #6797](#)

serialize execution witness with ssz instead of rkyv

move the serialization closer to [the specification](#)

state is stored as flat SSZ nodes and tree needs to be reconstructed in the guest

[ere #374](#)

add optimized llvm parameters for the zisk target

based on the paper [Evaluating Compiler Optimization Impacts on zkVM Performance](#)

also check my [write-up with findings](#)

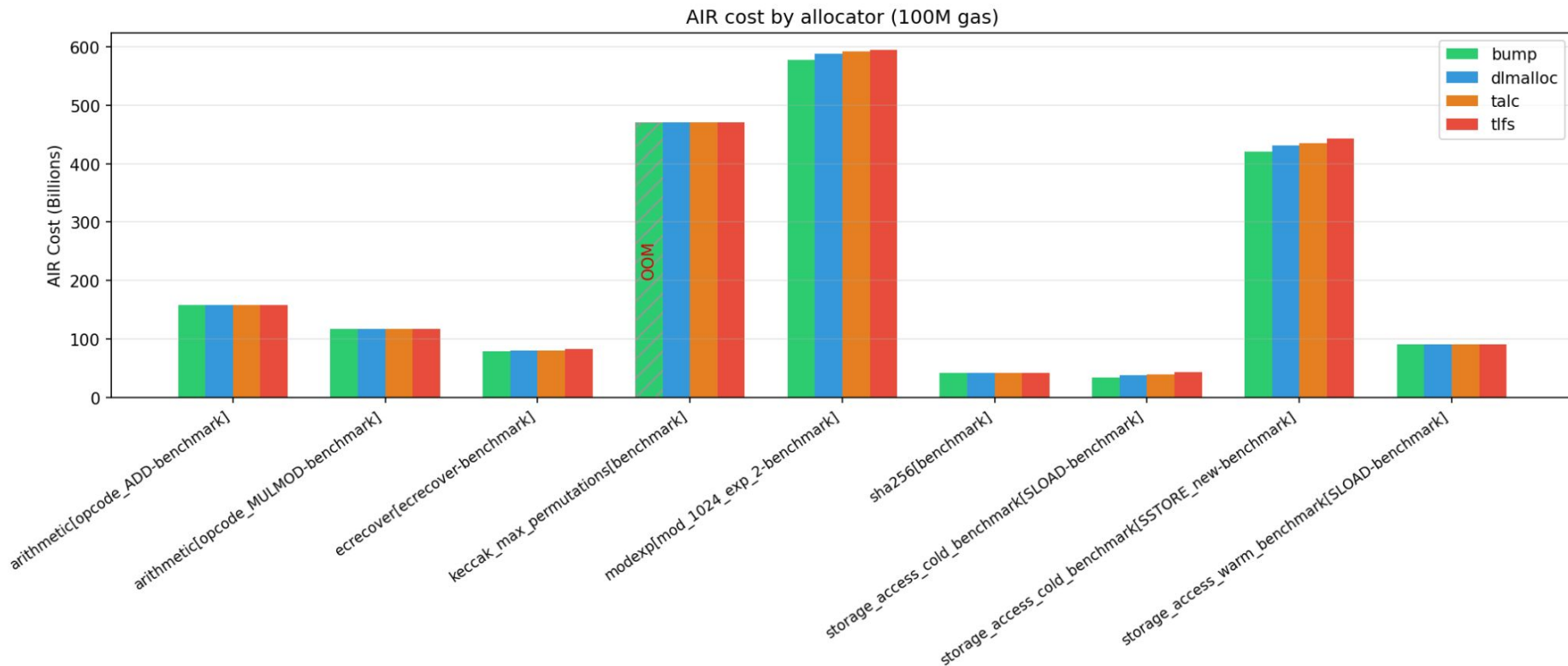
*requires no
source code changes*

allocator comparison in Zisk

bump is the fastest but how does it scale

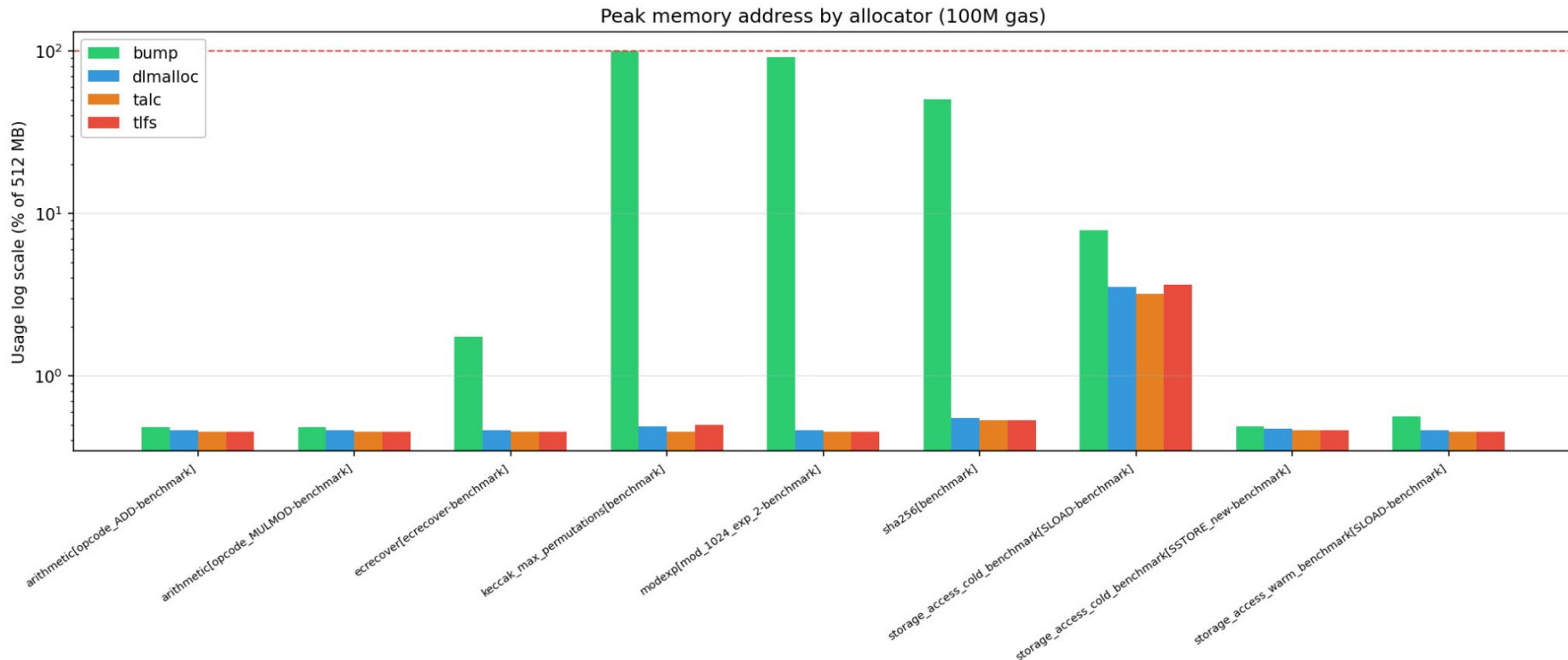
[write-up with findings](#)

100M EEST



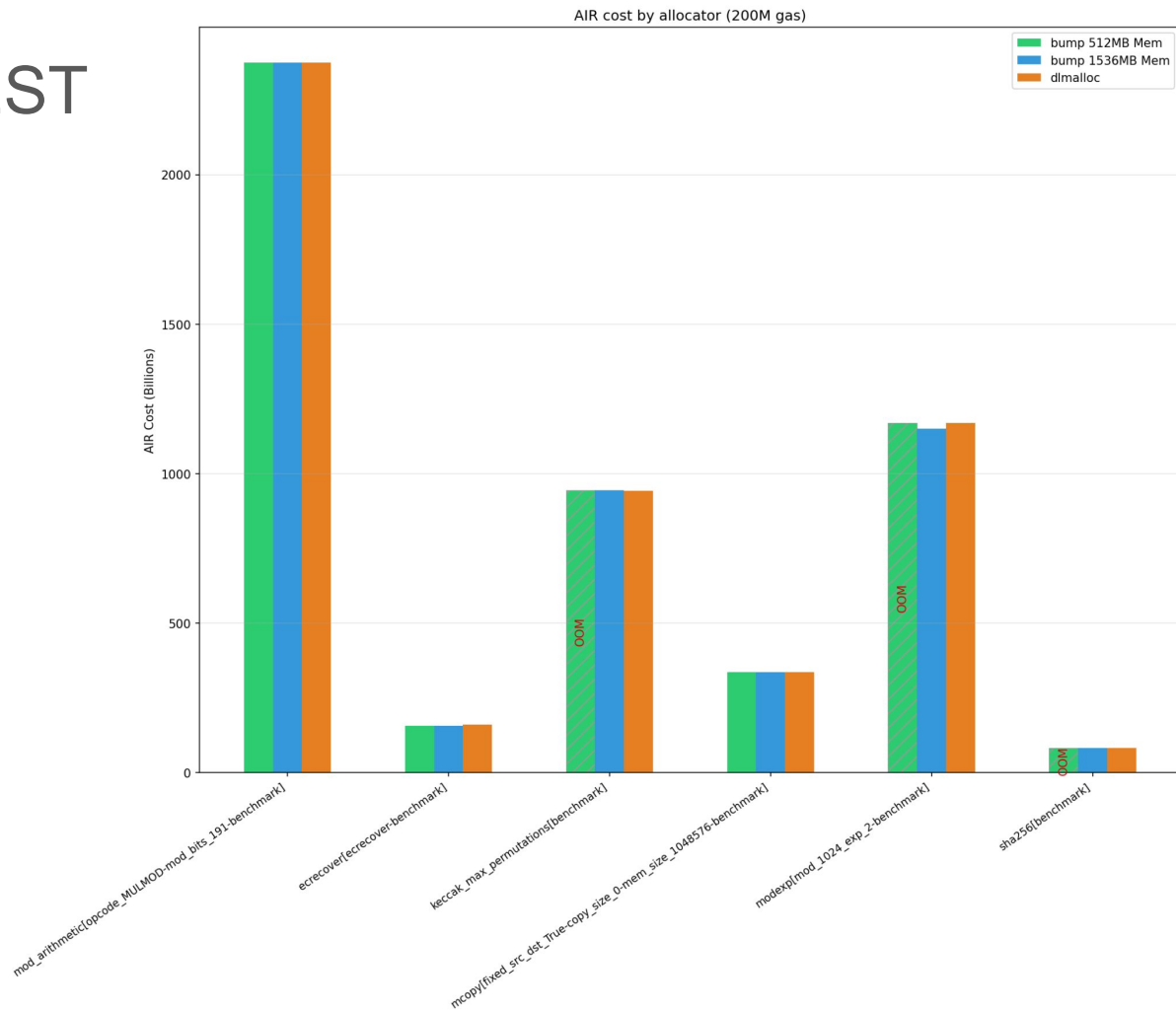
In all cases it holds: **bump** < **dlmalloc** < **talc** < **tifs**

100M EEST

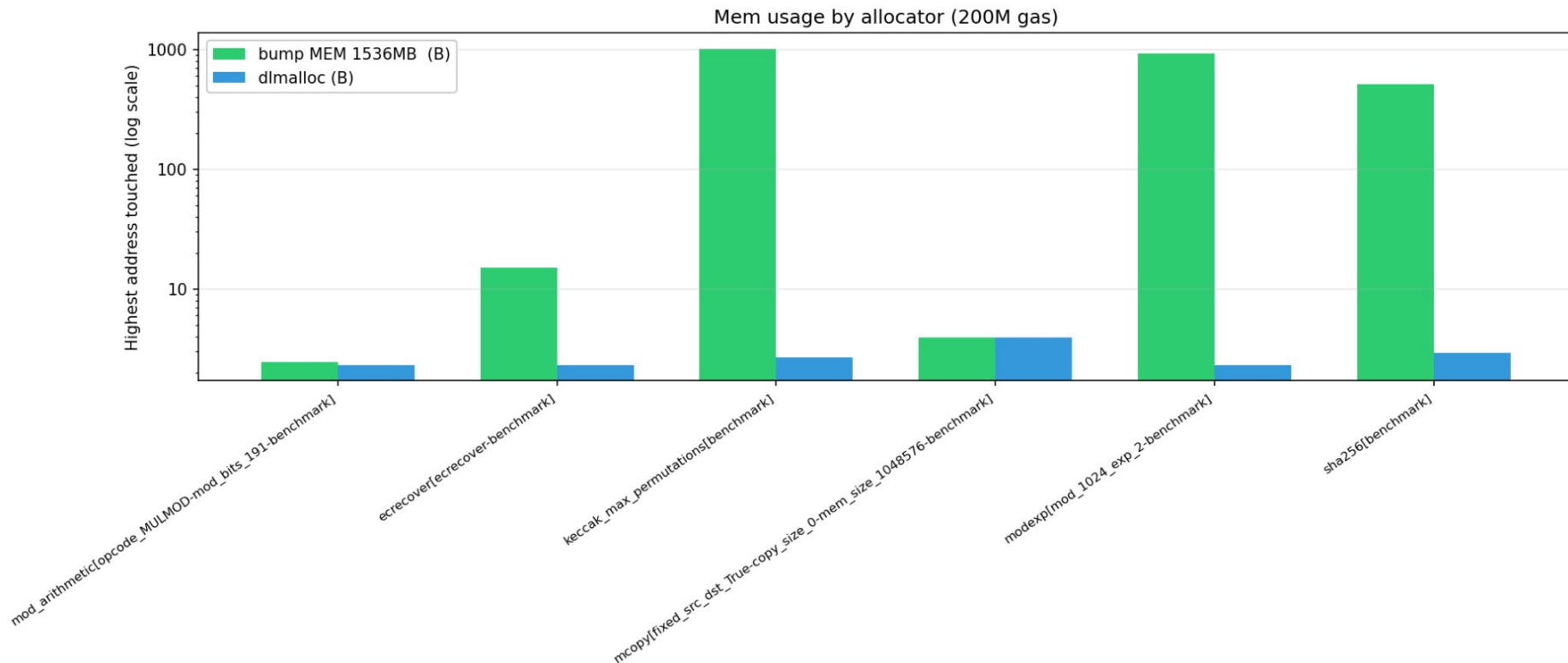


Unsurprisingly, **bump** consumes significantly more memory
danger of OOM

200M EEST



200M EEST



While **dlmalloc** introduces some overhead it is a safer option

Thanks zkEVM team for support<3

Mainnet blocks: 24949787 - 24949836	bump	d1malloc	talc	tlsf
Min gas block 14M: 24949797	24.71B	26.27B	26.52B	27.53B
Max gas block 56M: 24949791	77.71B	83.18B	83.86B	87.24B
Average over all blocks	45.44B	48.61B	49.02B	51.06B

Cost Breakdown (50-block average)

